

# One control core. Power conversion at every scale.

OpenVVF separates a reusable dual-MCU control and safety core from the power hardware. The same interfaces support smaller user-supplied stages, the 465 A-class Size 2 inverter, a released 15 kW DC/DC module, and a future 1,400 A chassis. Hardware, firmware, safety analysis, and test evidence are published openly.



The Size 2 control and power stack is built, bench-tested, and undergoing validation.

## 200-450 V

Size 2 DC-bus class by capacitor selection

## 465 A

Continuous design rating · 600 A peak (60 s)

## 10

Published test reports, raw telemetry linked

## PMSM + induction

Instrument-verified motor self-commissioning

### ■ Modular by design

The controller is largely **power-stage agnostic**. PWM and protection interfaces can connect to lower-power or user-supplied gate-drive stages with appropriate sensing and gate-drive scaling, while the same core extends to higher-voltage and higher-current hardware.

### ■ Built and bench-tested

Size 2 has completed a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and instrument-verified **motor self-commissioning**. Its dual-MCU architecture provides **six safe-state pathways**.

### ■ Open research at UCSC

OpenVVF is developed in the **Smart Power Lab** at UC Santa Cruz, with Professors **Keith Corzine** and **Leila Parsa** as faculty contacts. Project designs and **interactive telemetry logs** are public at [docs.openvvf.org](https://docs.openvvf.org).

## MODULAR POWER BLOCKS

Scale down or up around one control core

**CONTROL MODULE** Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



### Lower-power stages

Interface-ready for smaller or user-supplied gate-drive hardware

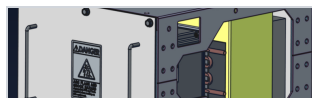
■ SCALABLE INTERFACE



### Size 2 inverter

200–450 V class · 465 A continuous design rating

■ IMPLEMENTED, UNDER TEST



### Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

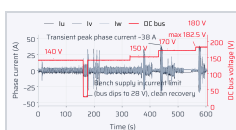
■ RELEASED DESIGN



### Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ IN DEVELOPMENT



## Live telemetry, not promises

Test reports include interactive logs for the browser-based Telemetry Viewer. Scan to open the 180 V bring-up: 140 to 180 V bus steps, approximately 38 A peak current, and clean recovery from a supply current-limit event.



## SUPPORT THE SMART POWER LAB

### Support the Smart Power Lab at UC Santa Cruz.

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## CURRENT SUPPORTERS

